



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Lab Experiment-1

Introduction to Assembly Language Programming

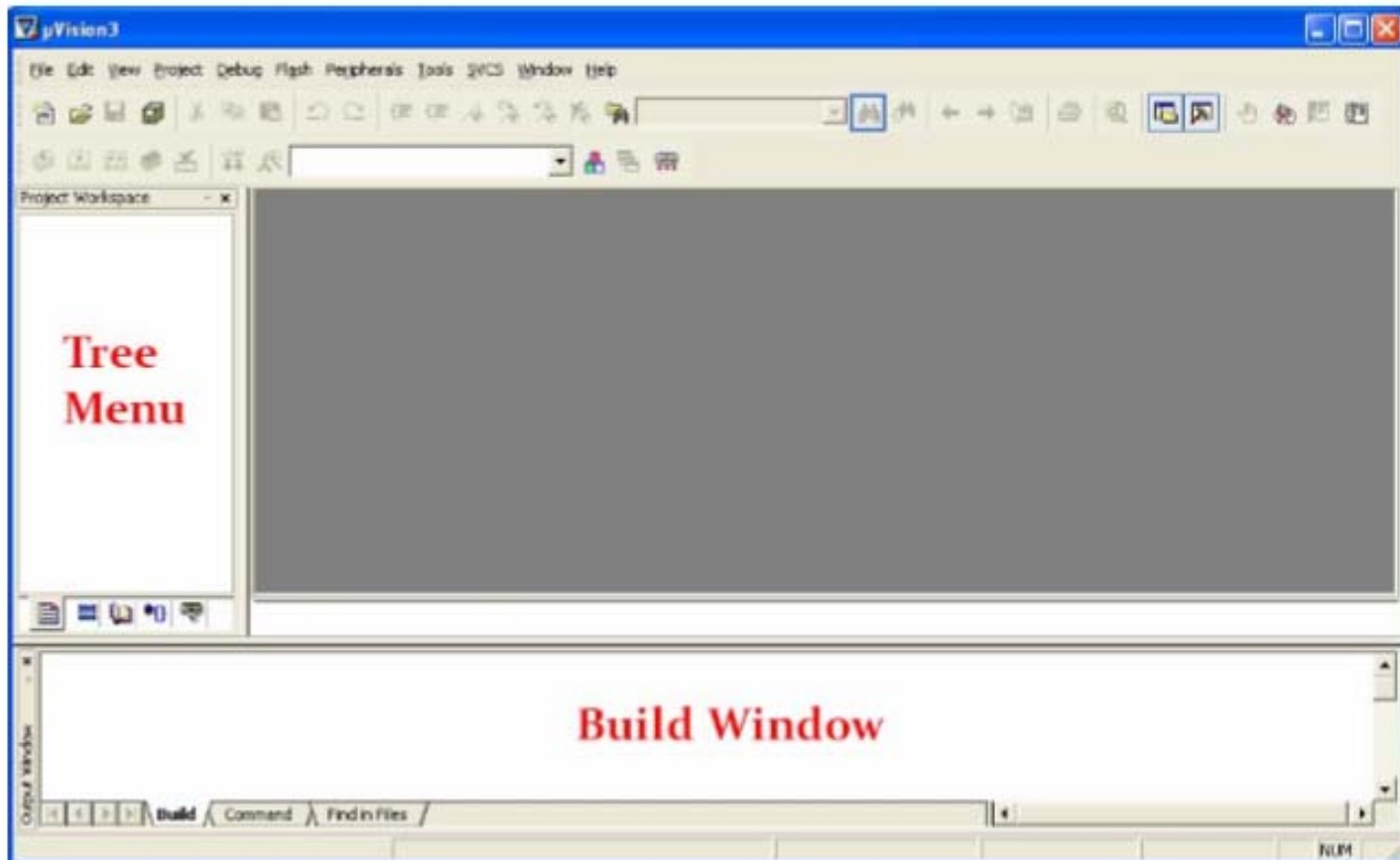
Dr. Sanjay Kumar Singh

Associate Professor Senior, Department of Communication Engineering,
School of Electronics Engineering

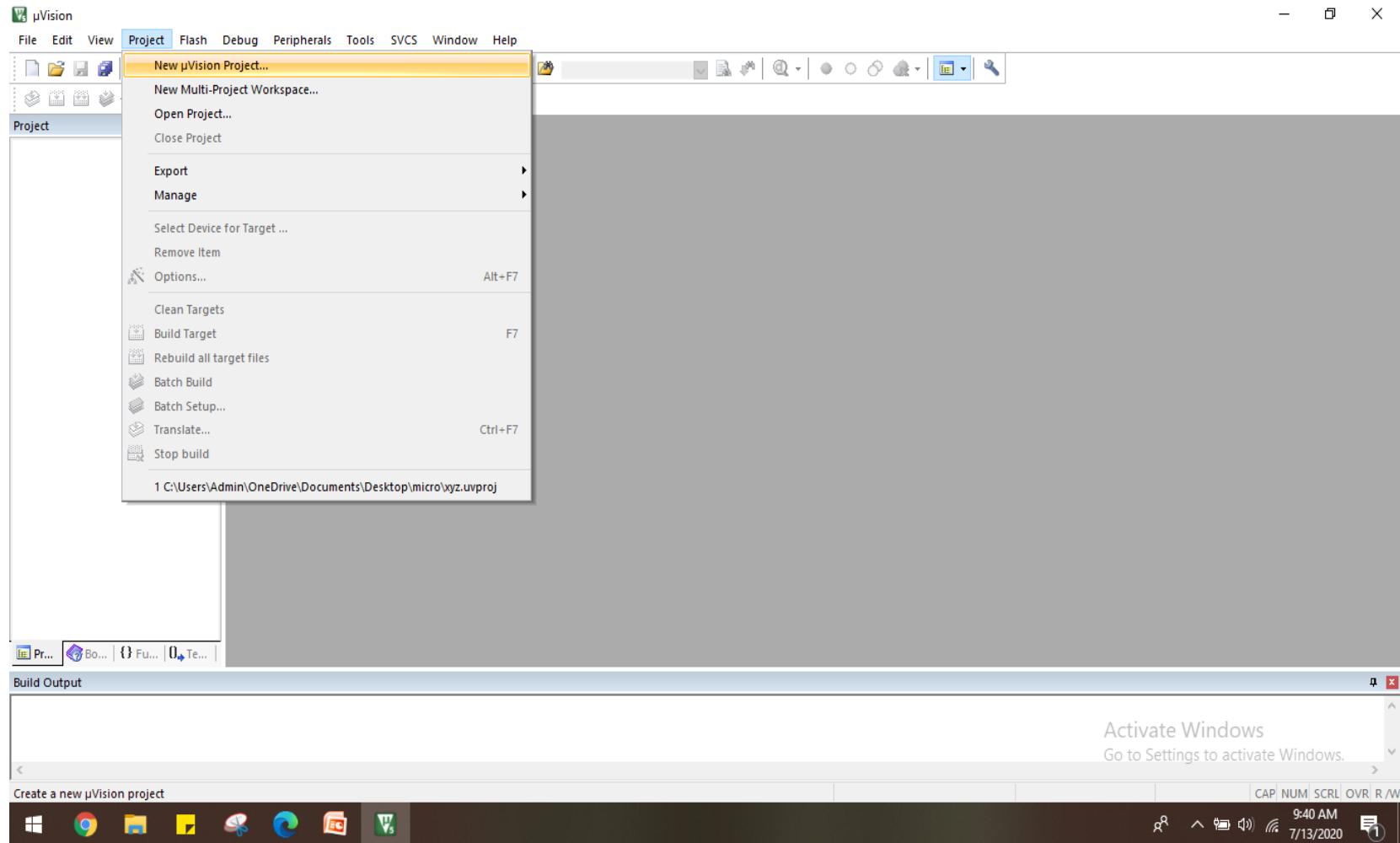
Winter 2025-26

Instructions to use KEIL SIMULATOR

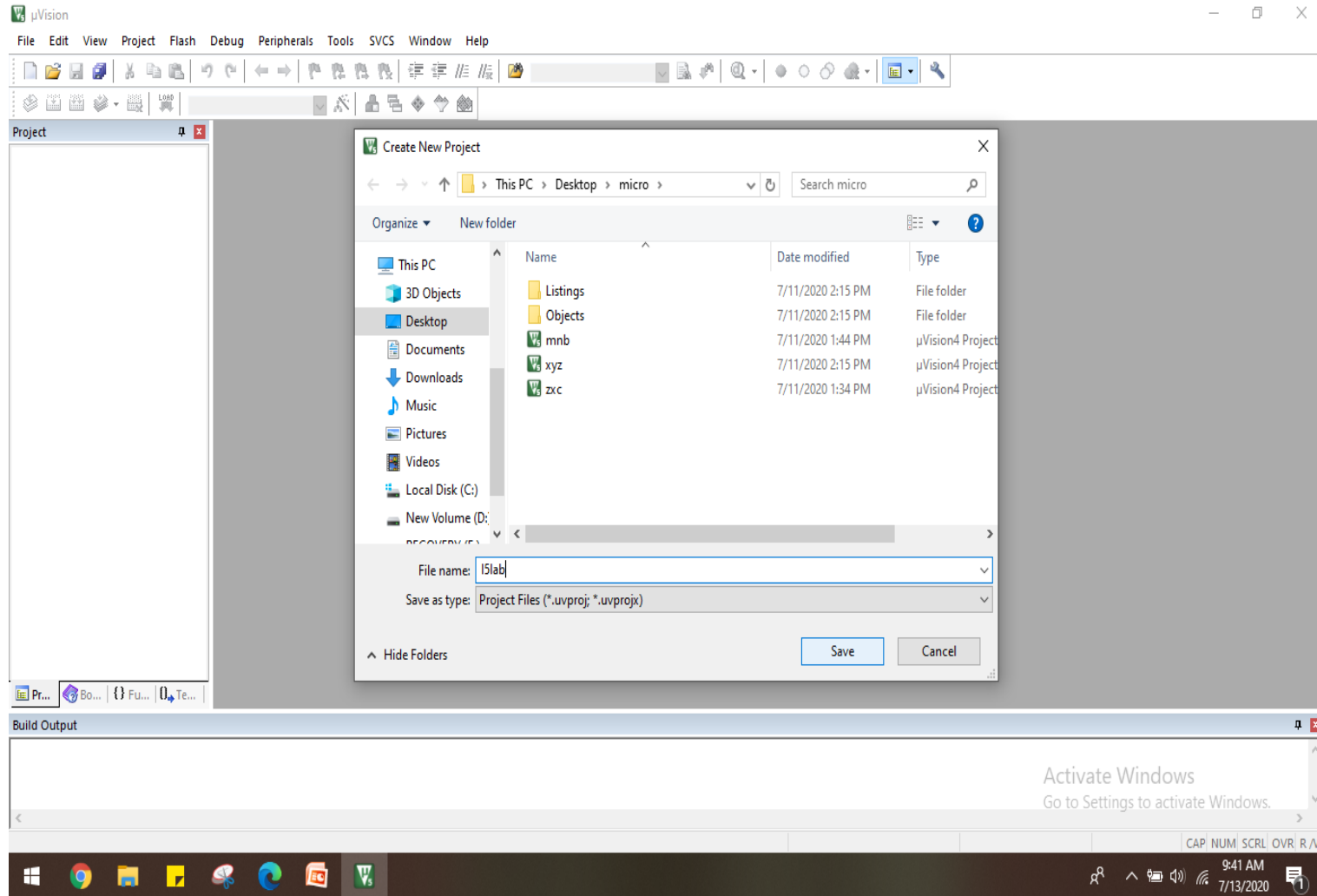
**STEP 1: Open Keil uVision5
(Start >Programs>Keil uVision 5)**



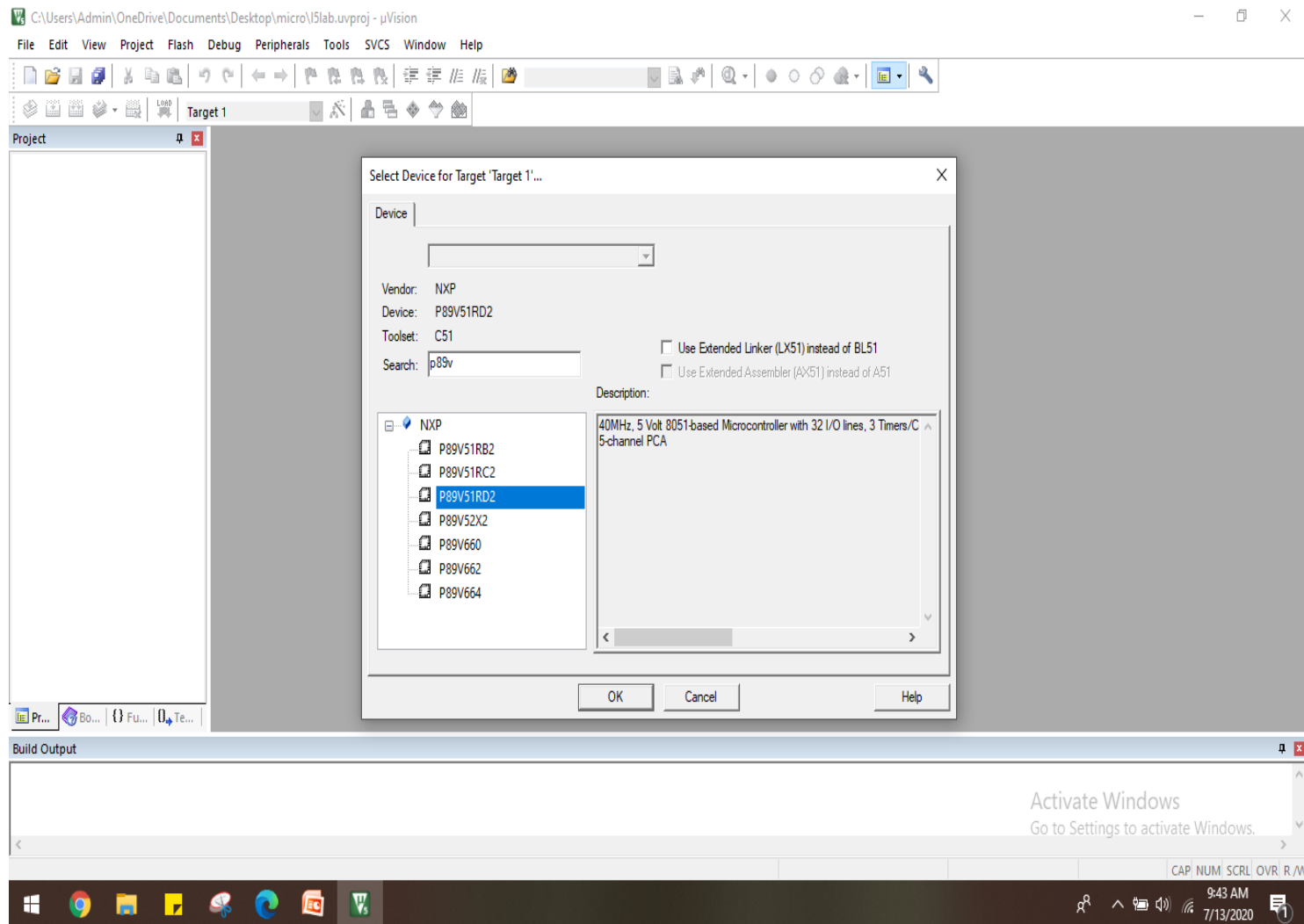
STEP2: Start a New Project (File>New>µvision project)



STEP3: Type any name without extension, and Save in your folder

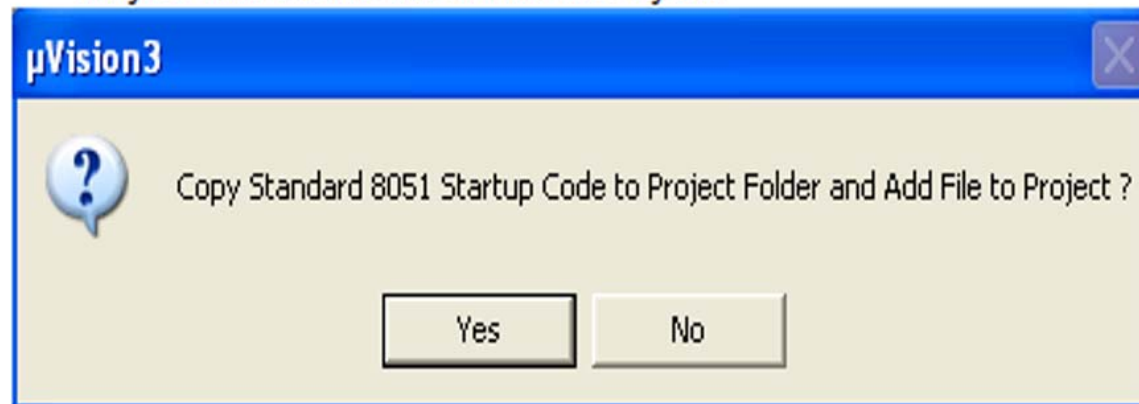


STEP 4: Select Nxp >P89V51RD2

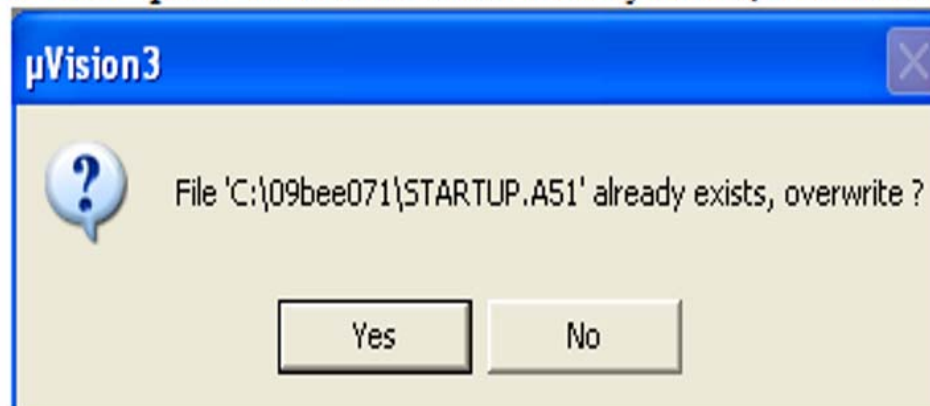


STEP 5 (Select→Yes)

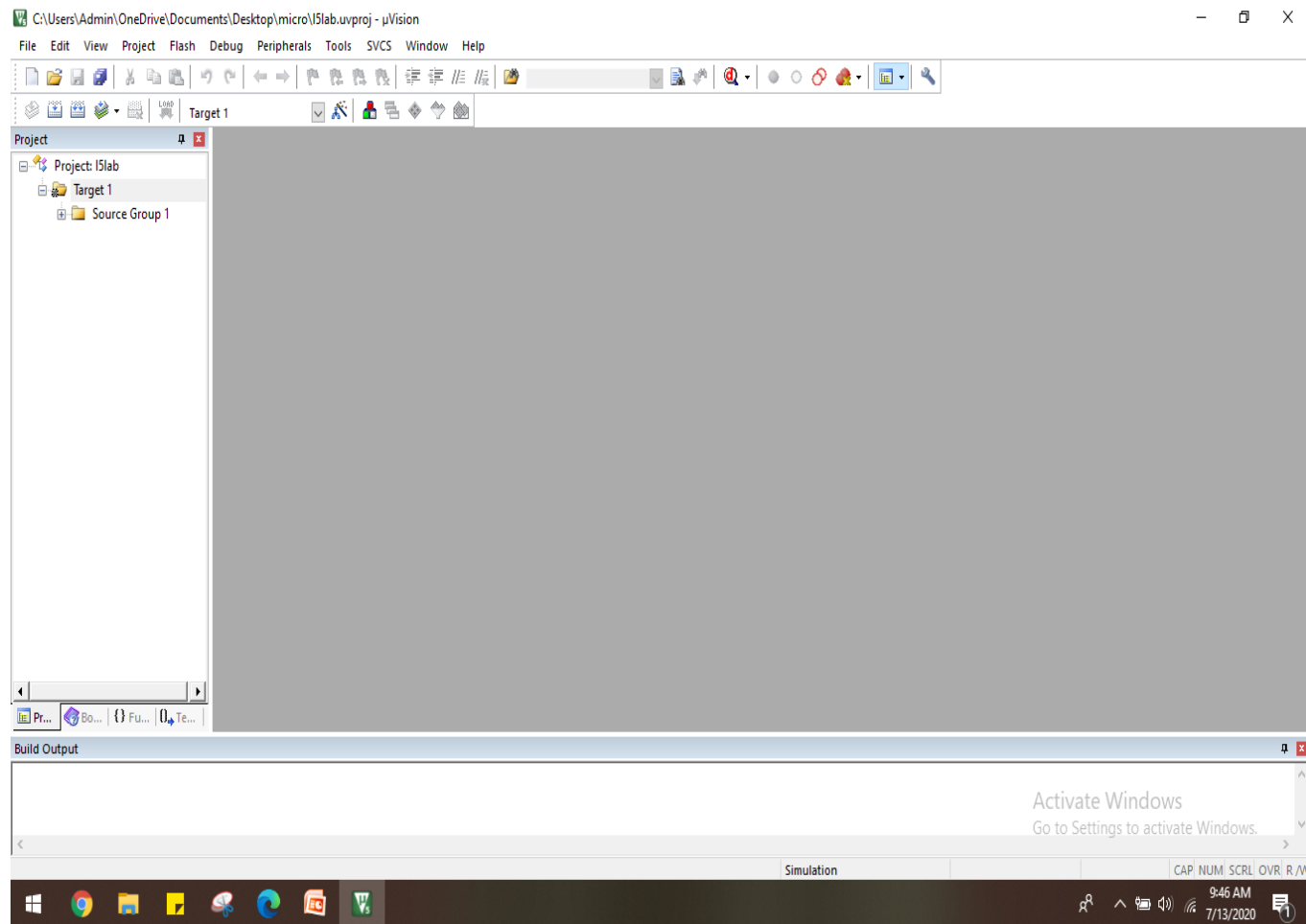
- Click Yes when asked whether to “Copy Standard 8051 Startup Code to Project Folder and Add File to Project.”



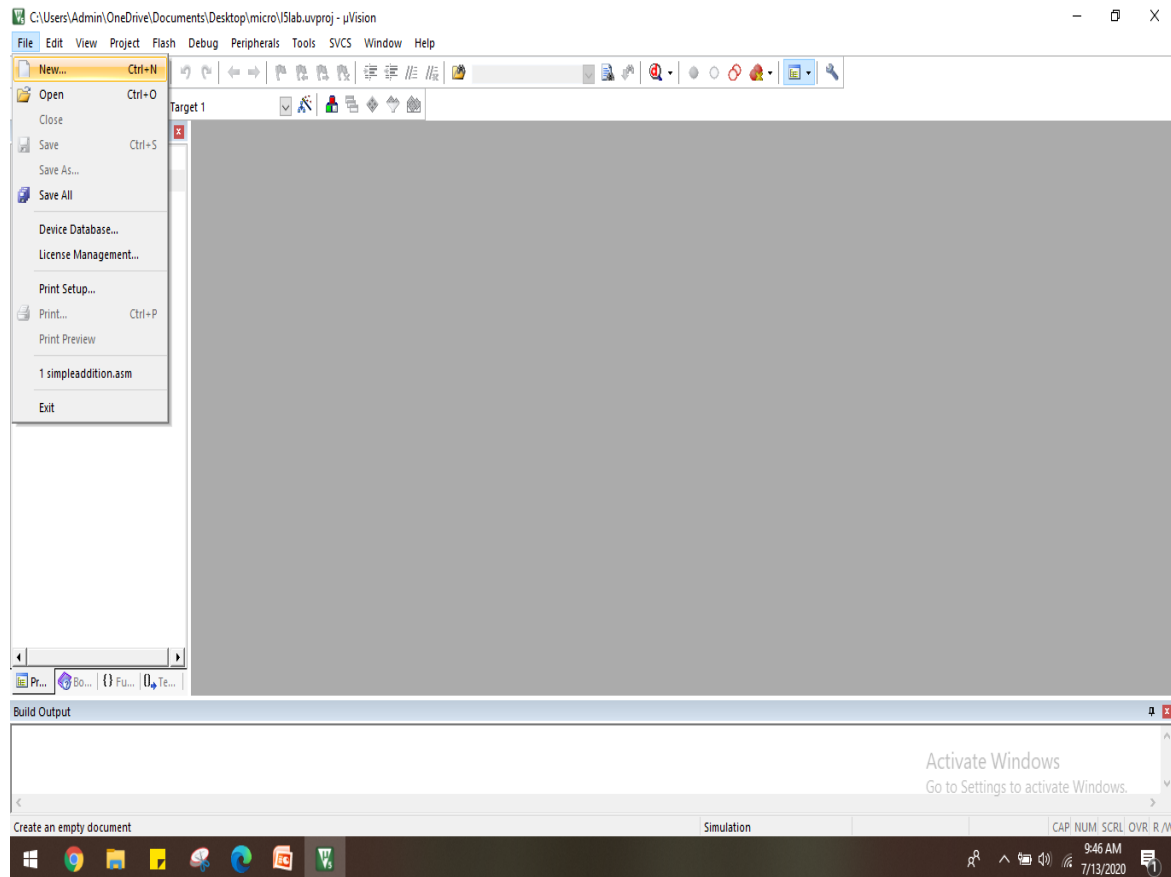
- Click Yes if asked “File <path>STARTUP.A51 already exists, overwrite?”



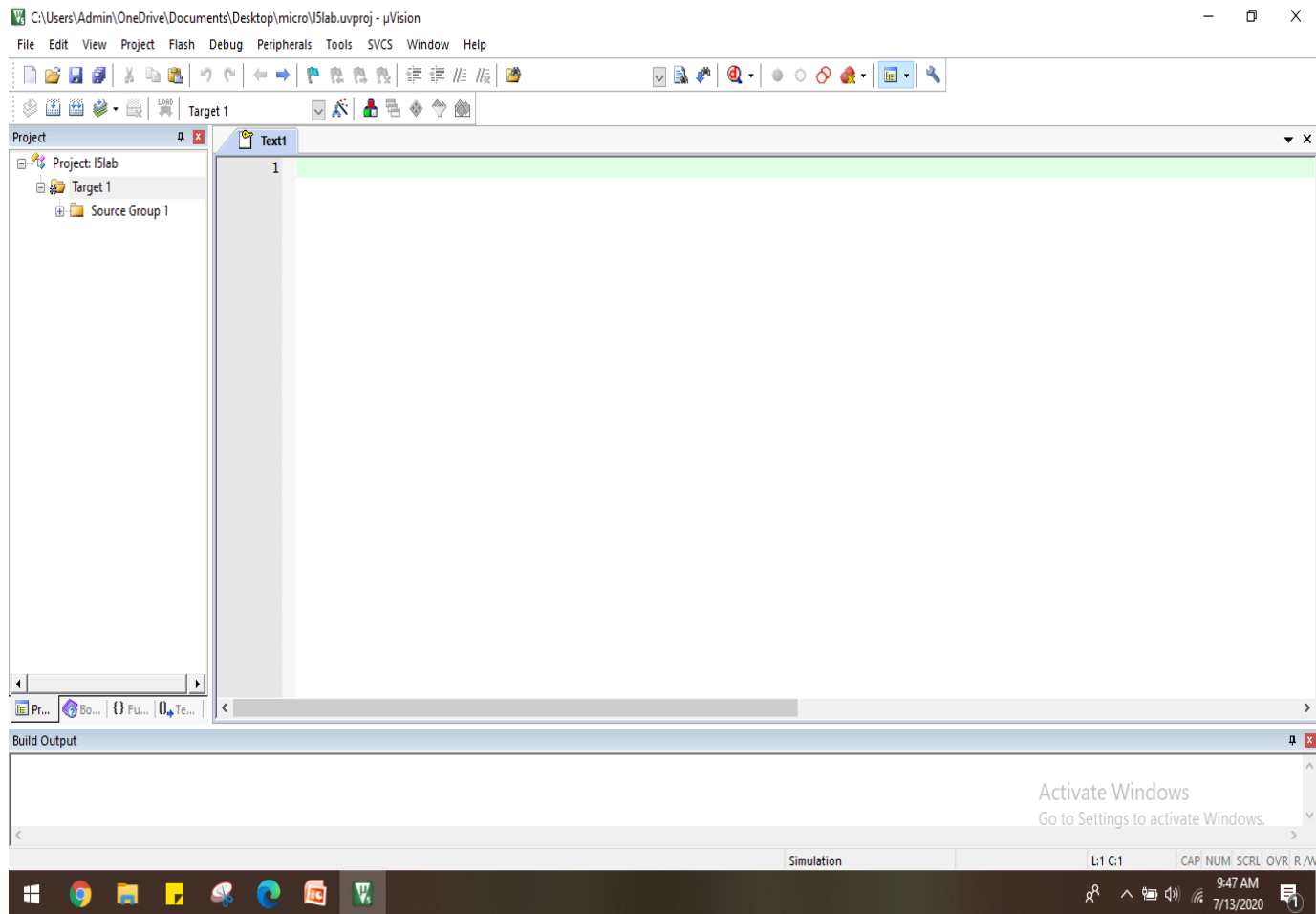
Blank space will appear



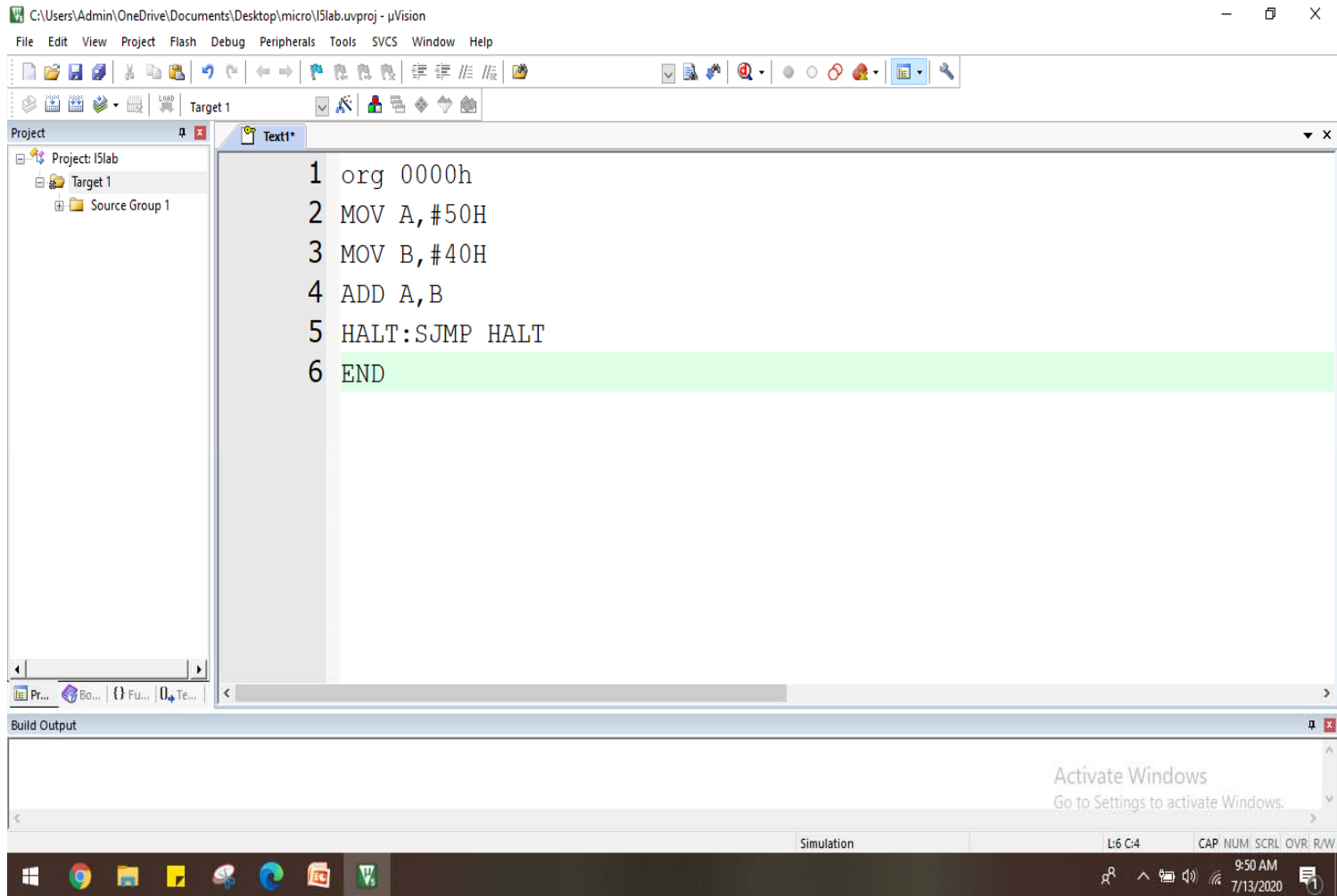
STEP 6: To create source file Open a new file (File→New File)



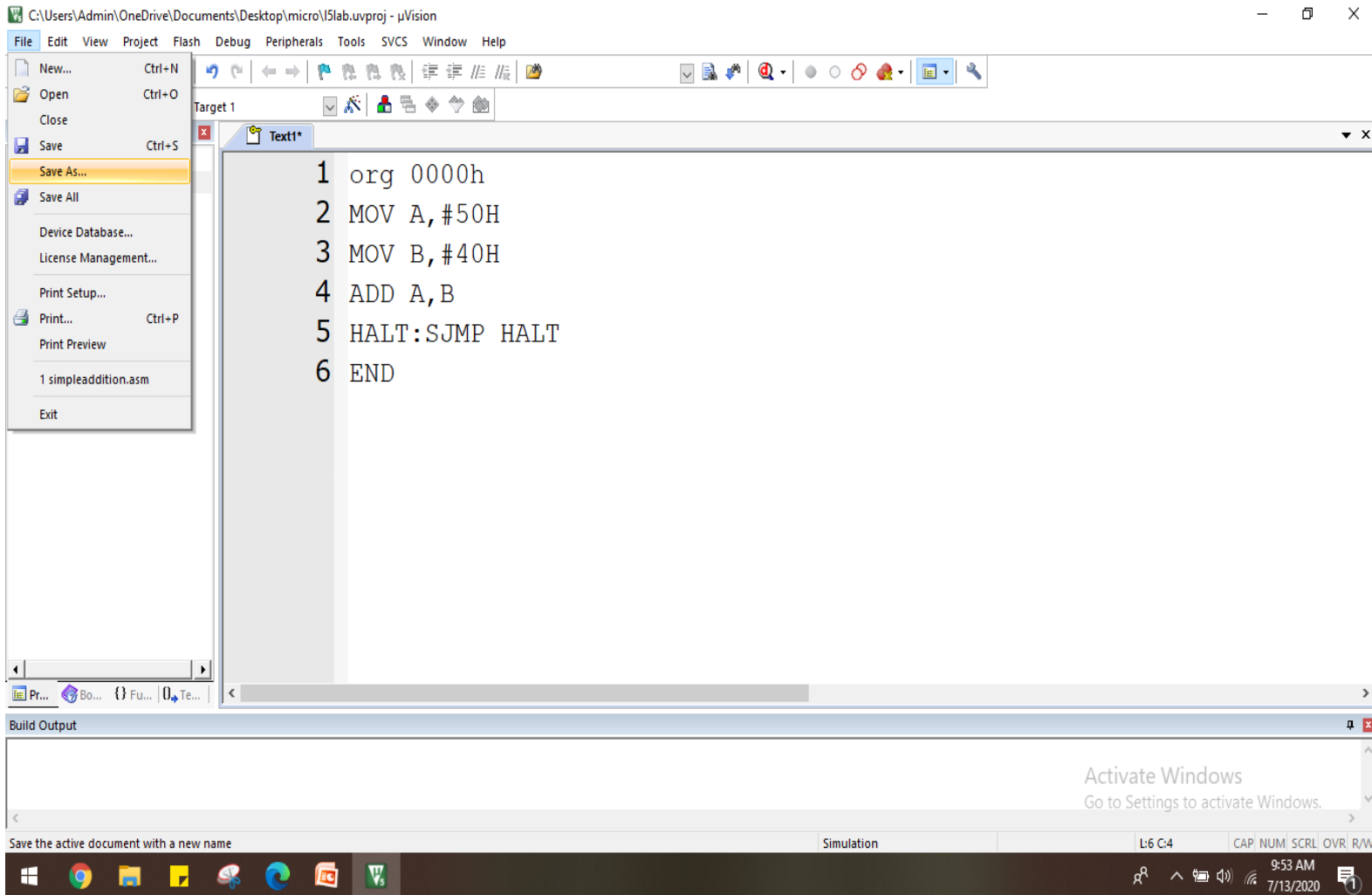
STEP 6 (Contd): A new window will open with blank space to type the code



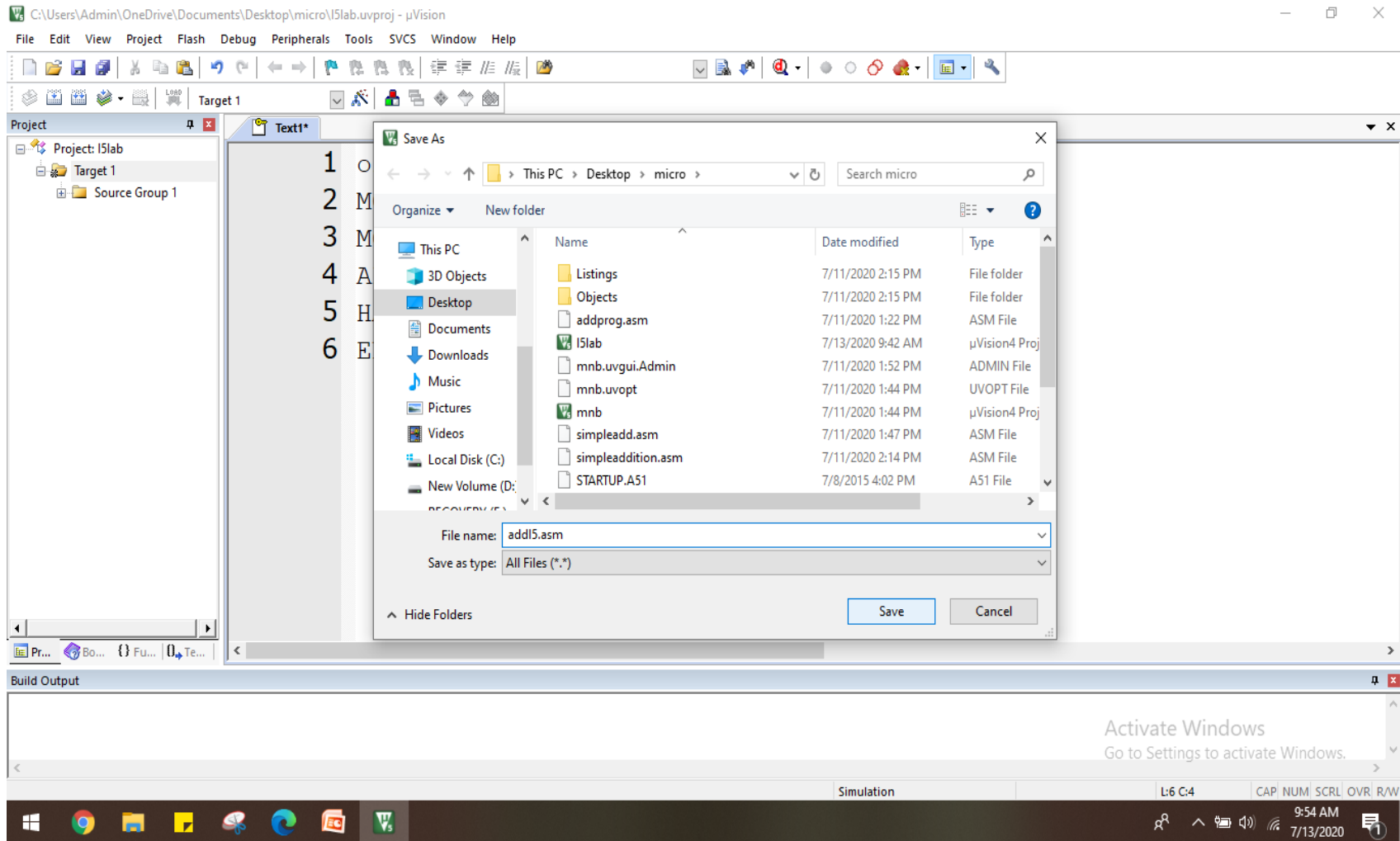
STEP 7: Type the Code



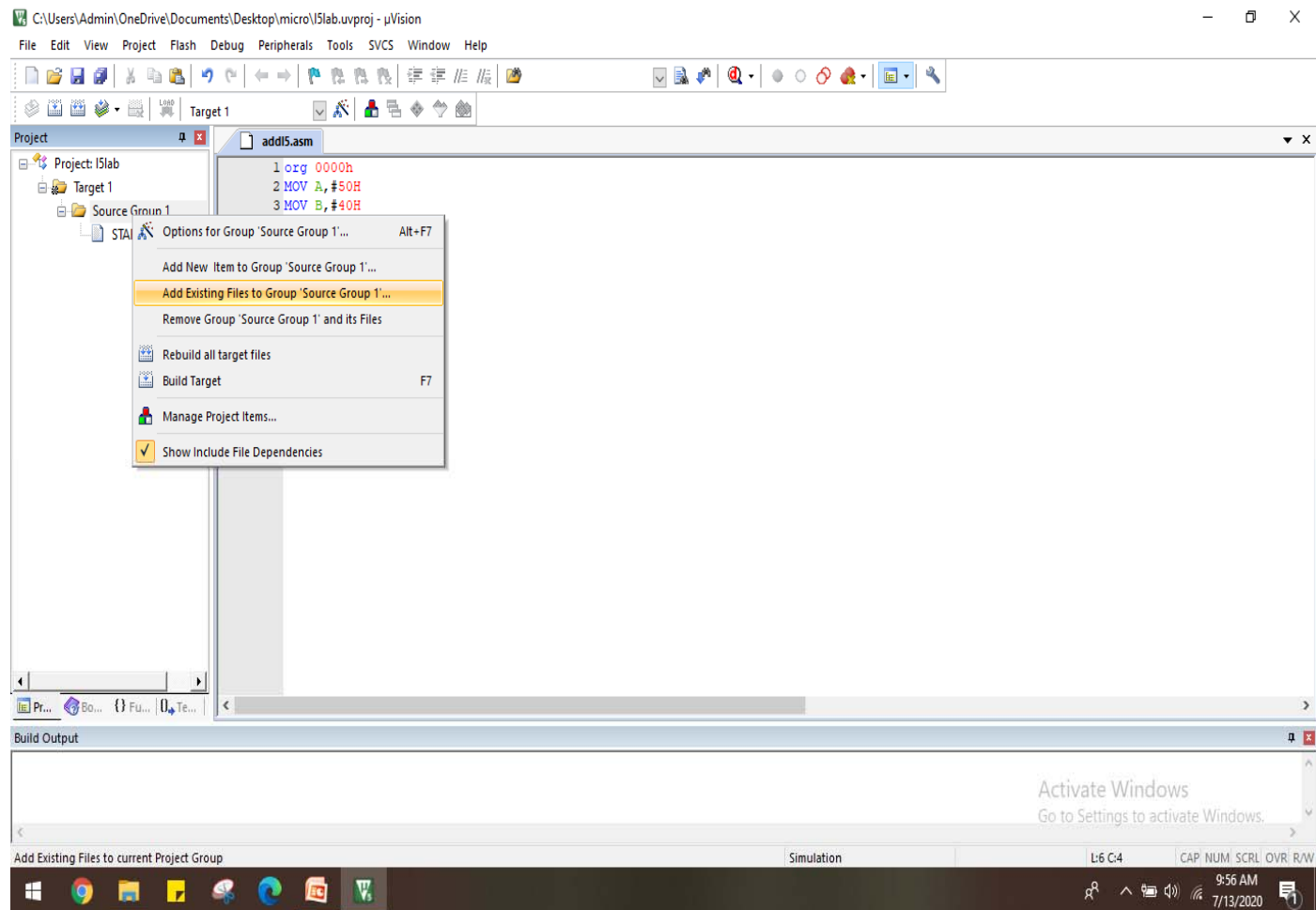
STEP 8: Click file menu and save



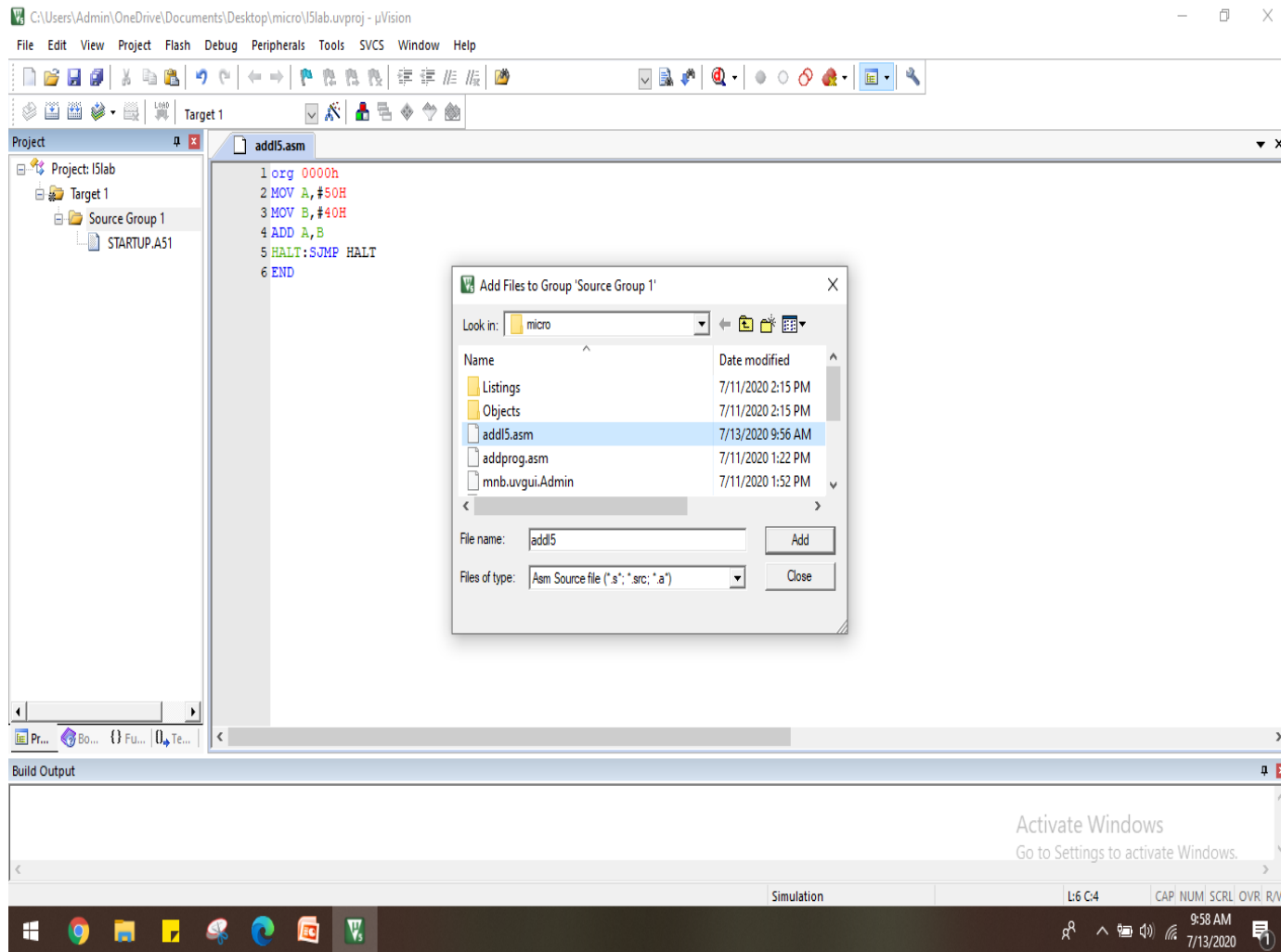
STEP 9: Save the file name with extension as → .asm



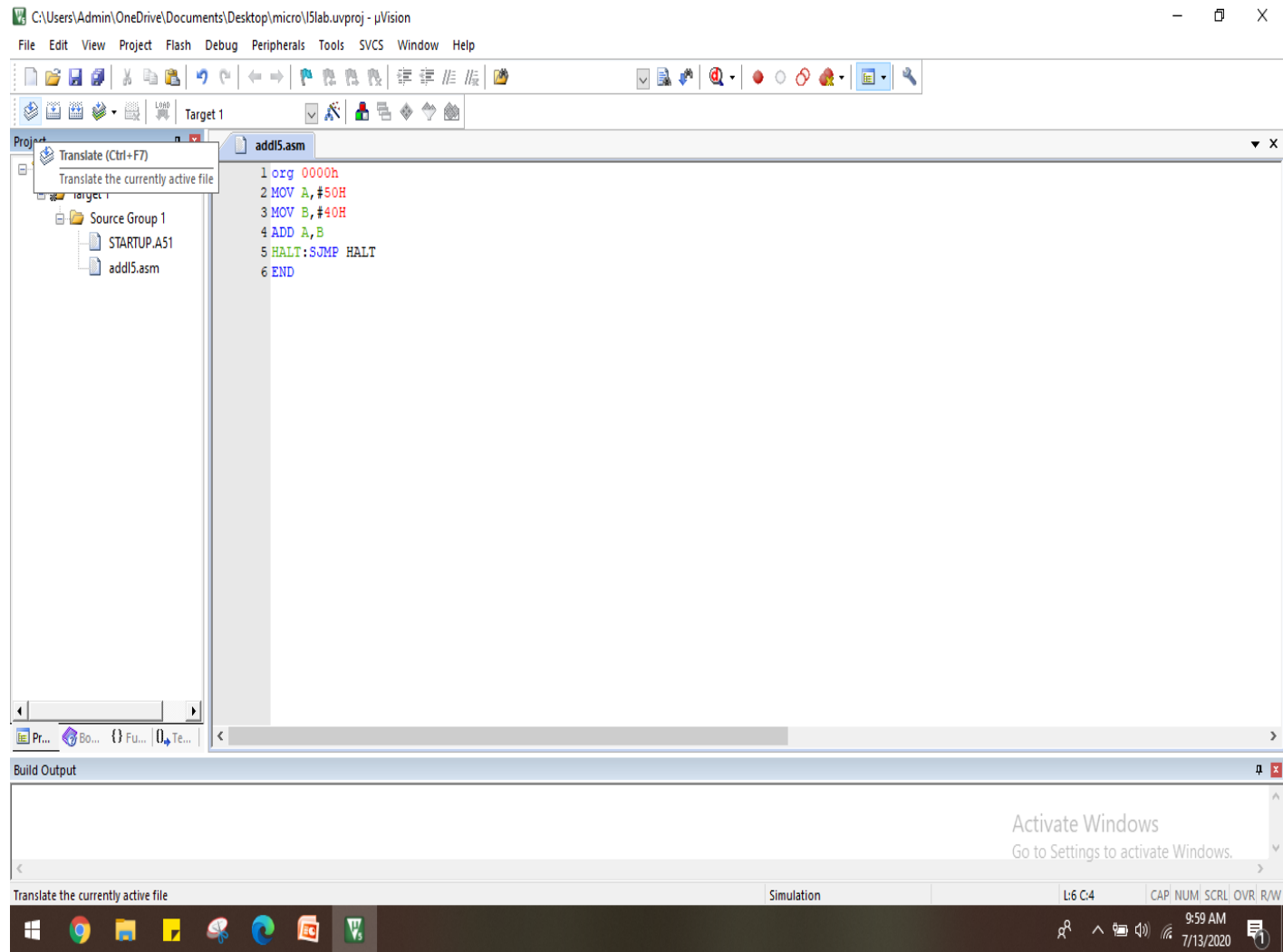
Step 10: Right click the source group and select→Add existing file to the source group



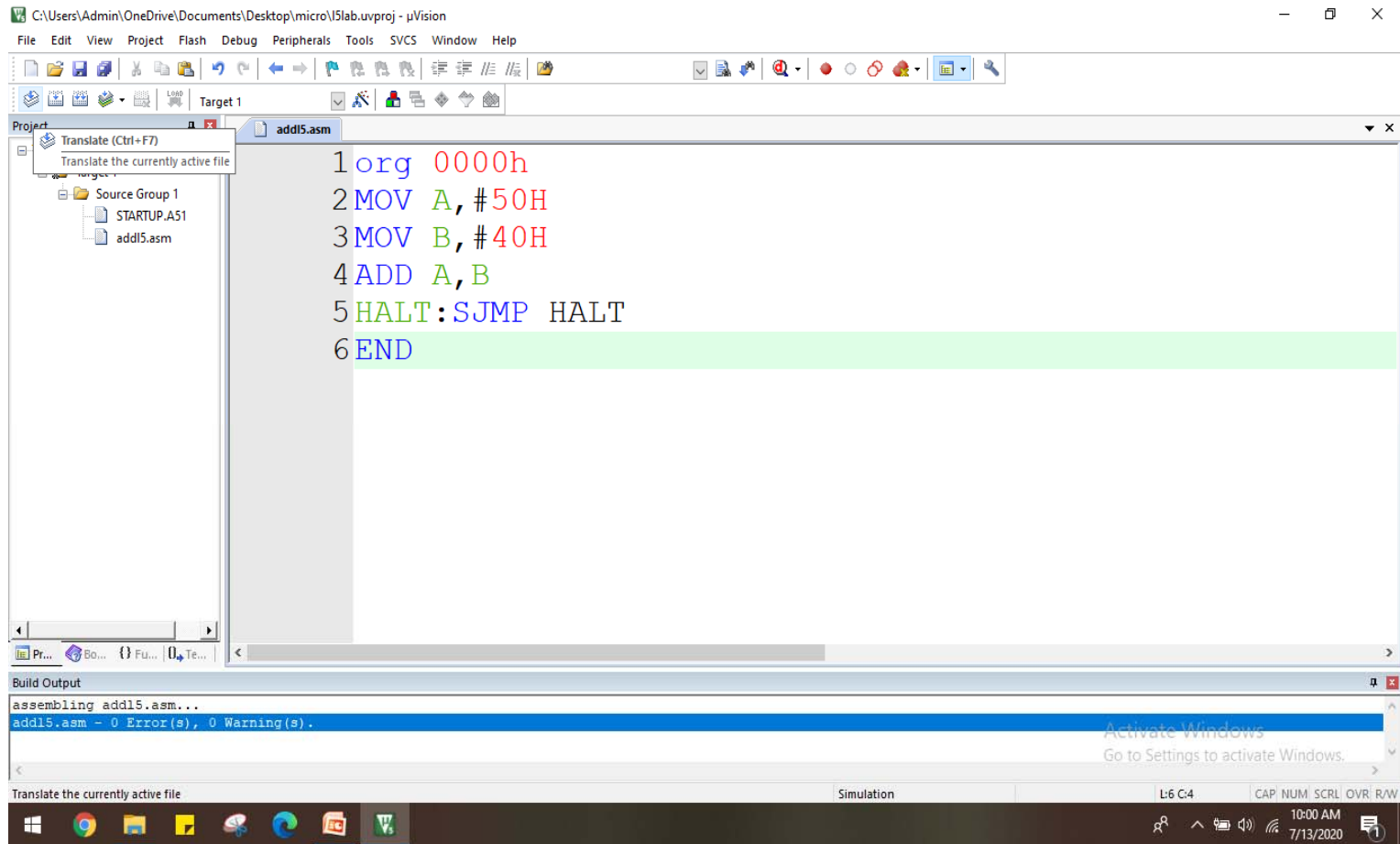
Step 11: select the saved file from the folder and click→ADD and close



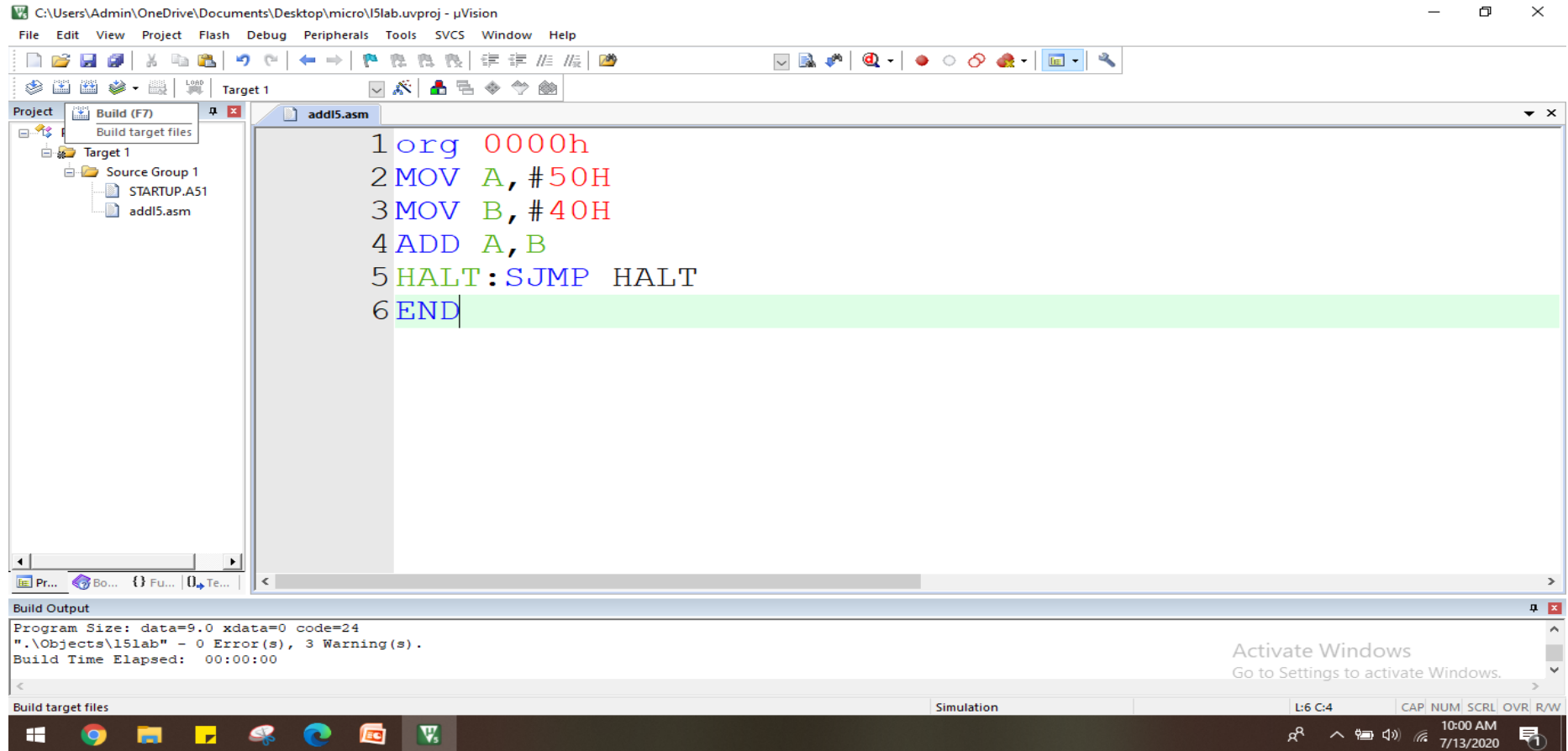
Step12: Expand the source group in project work space and ensure your file is under source group



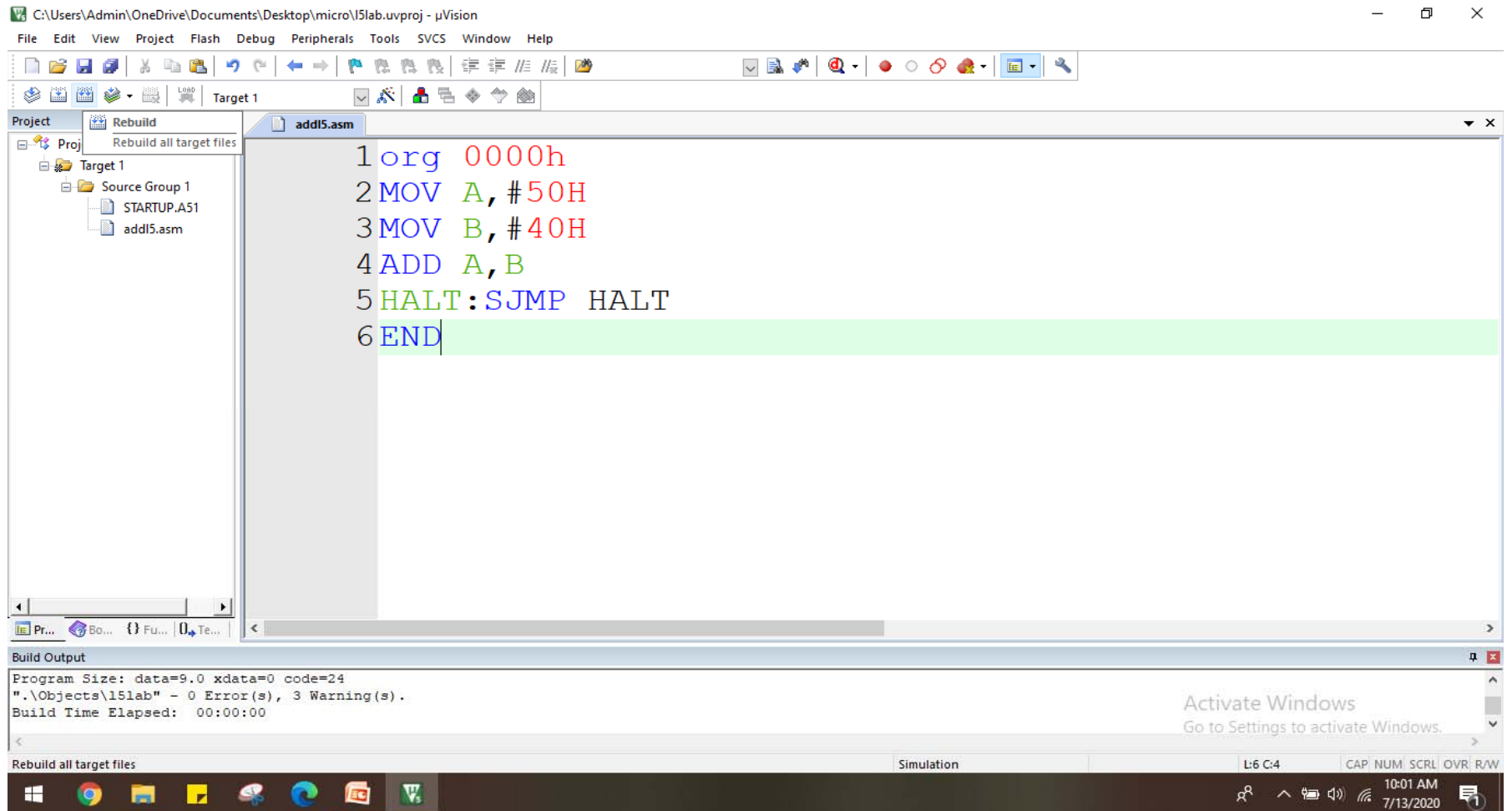
Step13: Select→”Translate” on the left corner and check for syntax Errors



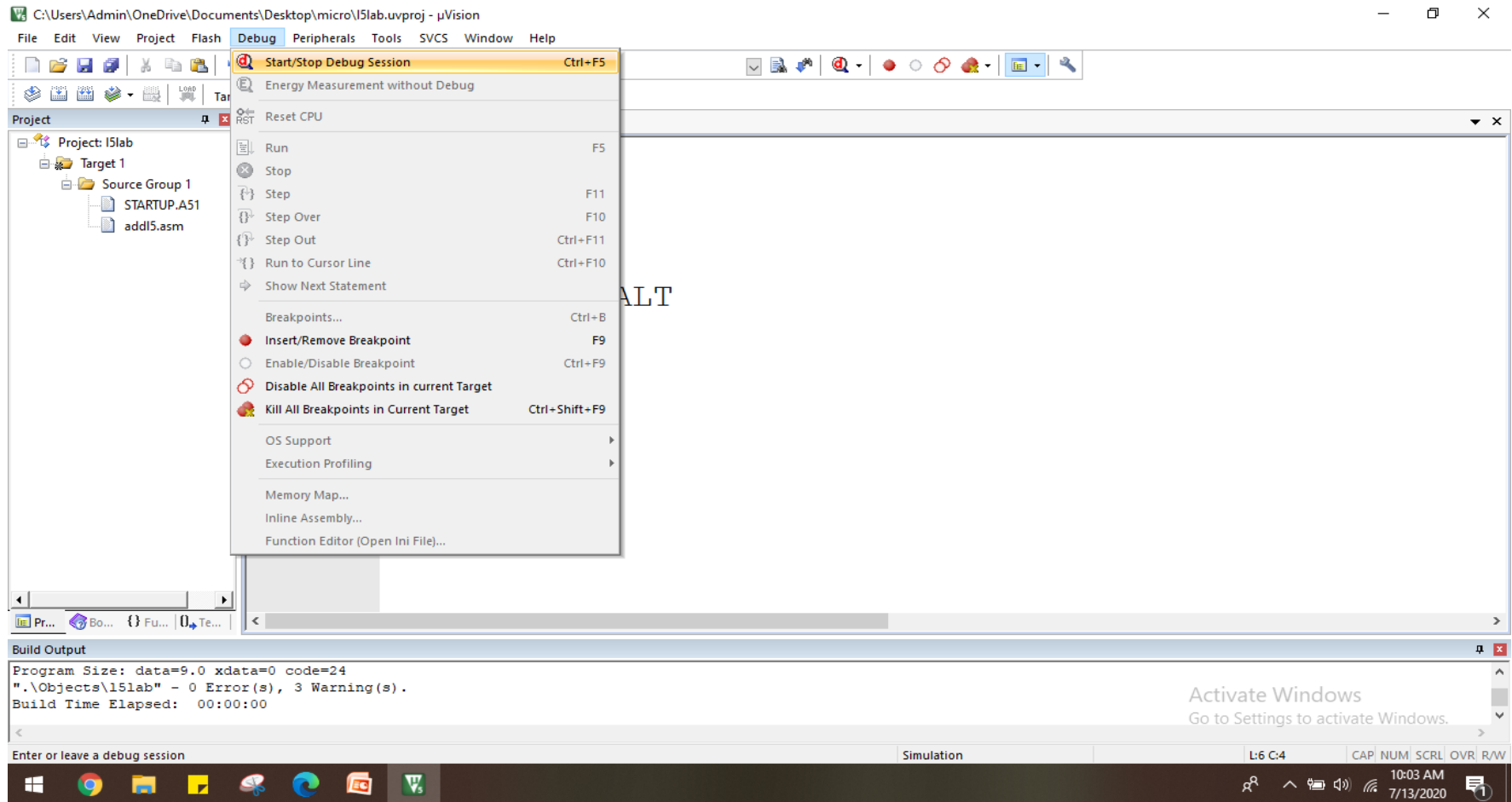
Step14: Select→"Build" on the left corner



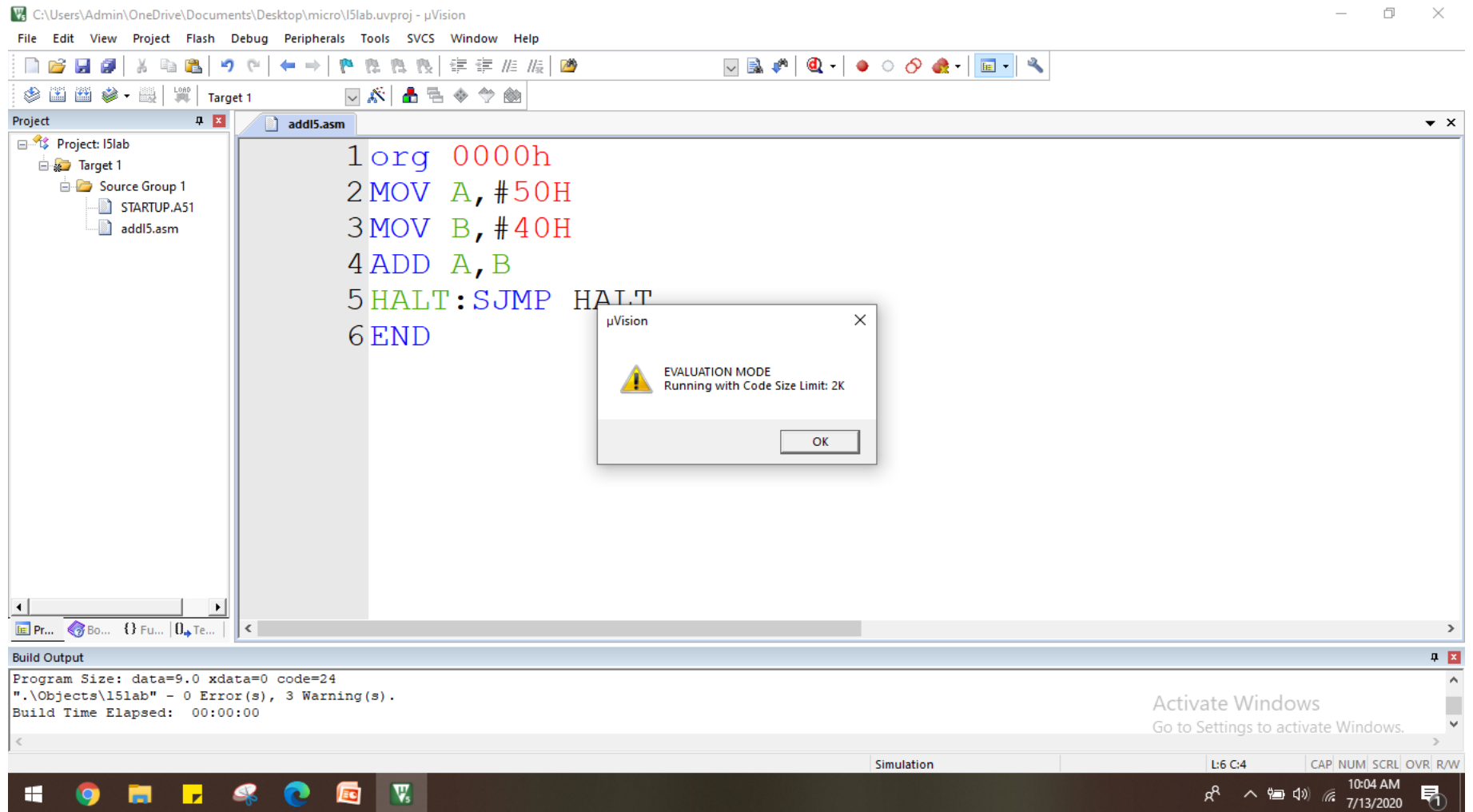
Step15: Select→”Rebulid” on the left corner



Step16: Debug→Start Debug



Step17: Click →"ok"



Registers will appear on the left

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
a	0x00
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0000
PC	0x0000
states	0
sec	0.00000000
psw	0x00

Disassembly

```
C:0x0000 7450 MOV A, #0x50
3: MOV B, #40H
C:0x0002 75F040 MOV B(0xF0), #0x40
4: ADD A, B
```

add15.asm

```
1 org 0000h
2 MOV A, #50H
3 MOV B, #40H
4 ADD A, B
5 HALT:JMP HALT
6 END
```

Command

```
Running with Code Size Limit: 2K
Load "C:\\Users\\Admin\\OneDrive\\Documents\\Desktop\\micro\\Objects\\15lab"
```

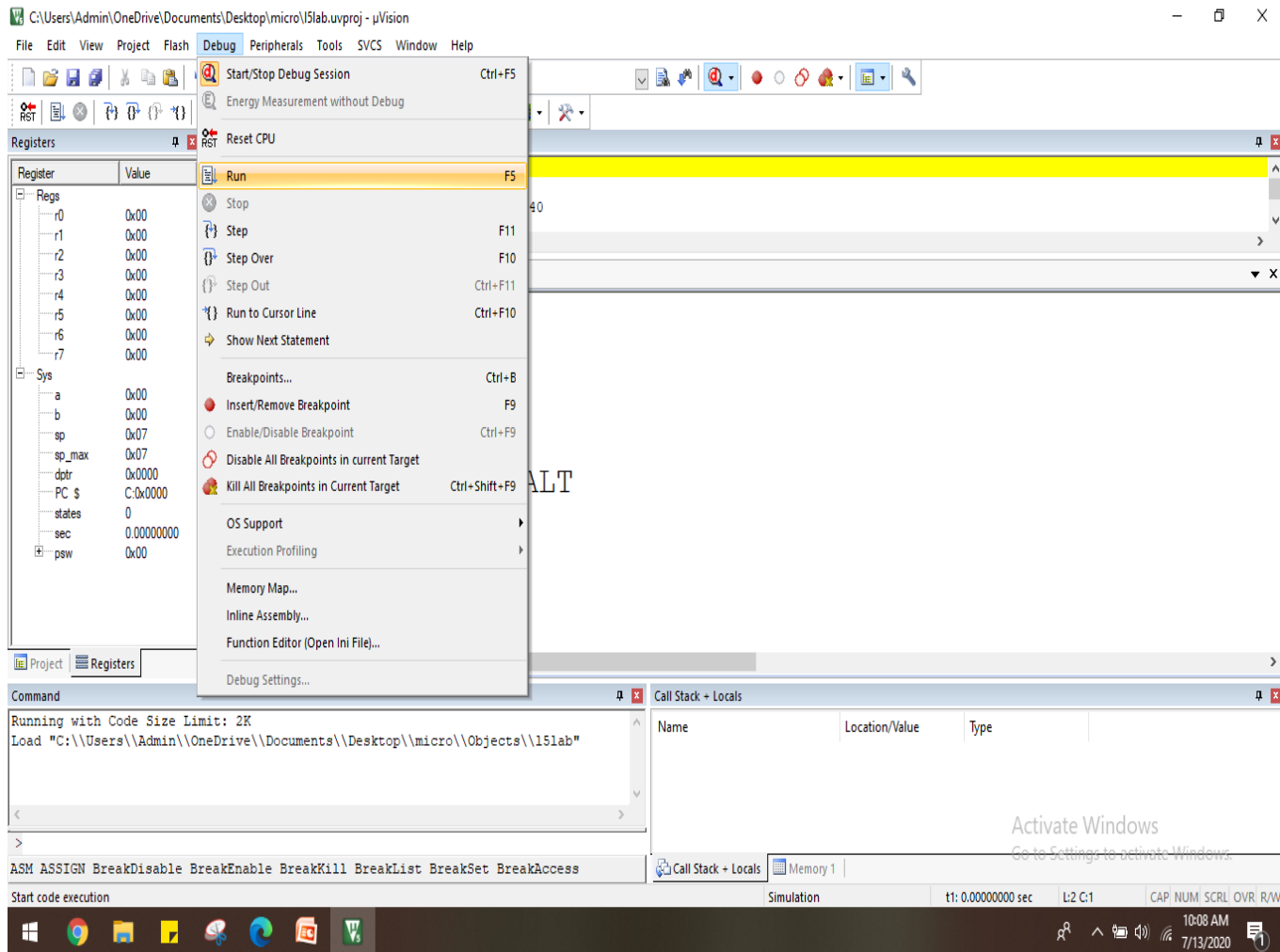
Call Stack + Locals

Name	Location/Value	Type
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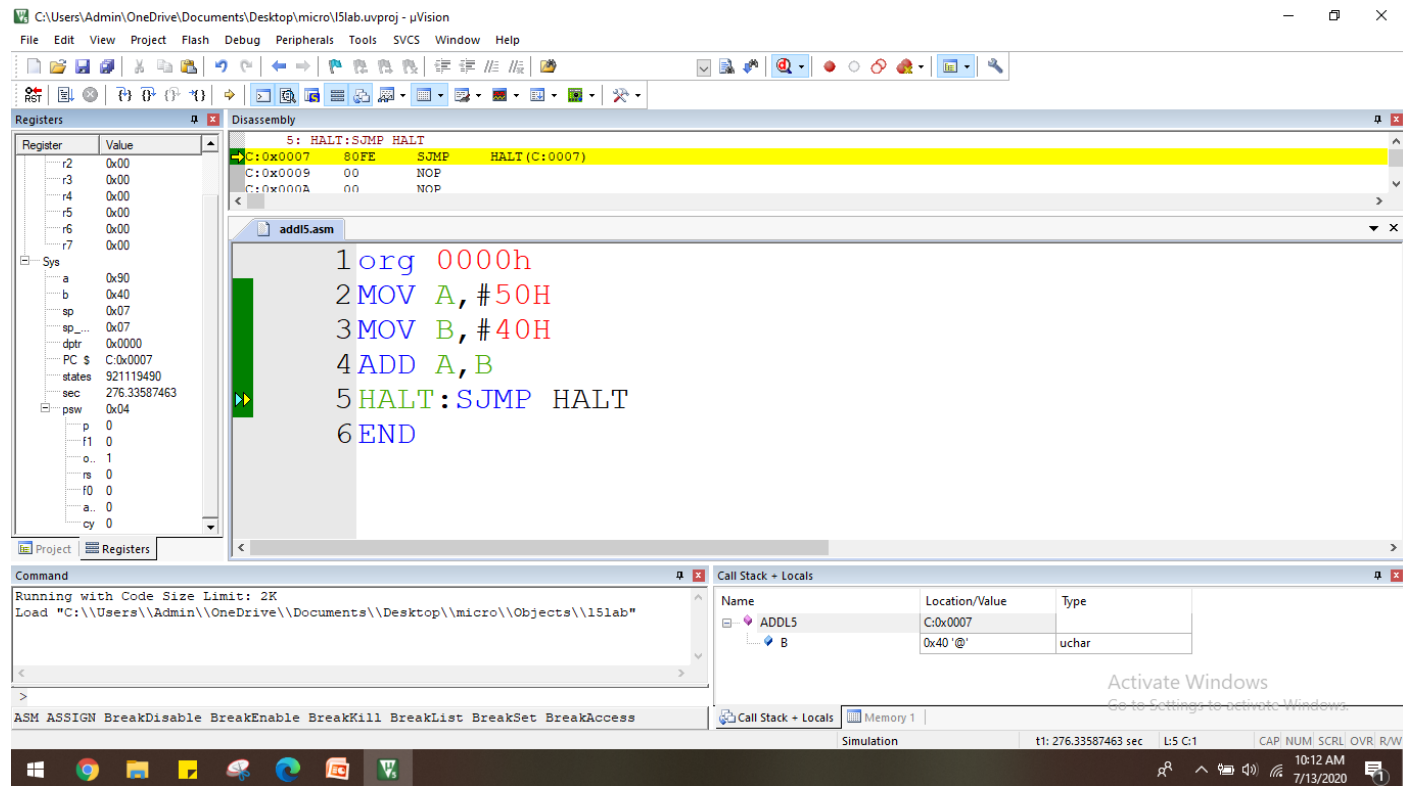
Simulation t1: 0.00000000 sec L:2 C:1 CAP NUM SCRL OVR R/W

10:04 AM 7/13/2020

Step18: Click Debug→Run



Check the results



TASK1-EXPT1: Perform the basic arithmetic operation using Keil micro vision

PROGRAM 1-ADDITION

```
ORG 0000H
MOV A, #05H
MOV R0, #05H
ADD A, R0
H: SJMP H
END
```

PROGRAM 2-SUBTRACTION

```
ORG 0000H
MOV A, #0AH
MOV R0, #05H
SUBB A, R0
H: SJMP H
END
```

PROGRAM 3-MULTIPLICATION

```
ORG 0000H
MOV A, #05H
MOV B, #05H
MUL AB
H: SJMP H
END
```

PROGRAM 4-DIVISION

```
ORG 0000H
MOV A, #20H
MOV B, #05H
DIV AB
H: SJMP H
END
```